

Generalizing the Augot-Finiasz PKE to Other Code Classes

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Abstract. The Augot-Finiasz system is a public-key encryption (PKE) scheme based on Reed-Solomon codes and was later followed by analogous versions in the rank metric. Although these schemes were eventually broken, their fundamental idea remains exciting. Notably, these schemes are significantly different from the McEliece system as there is no need to hide the code and, as such, promise much better parameters. Further, they admit a simple description where both the public key and ciphertext are just corrupted codewords of a public code. An interesting question is whether the general idea can be made to work, i.e., resist all known attacks, by using other code classes. This paper shows how to generalize the Augot-Finiasz system to other code families. We reduce the correctness and security of this framework to simple assertions about the code class with which it is instantiated. Specifically, its correctness is equivalent to the existence of an efficient error-erasure decoder, and its security reduces to an easily understood hardness assumption, called *supercode decoding*, close to the syndrome decoding problem. We provide a negative answer for various code families by showing that solving the supercode decoding problem is easy for them. It remains an open question whether a secure choice exists.

Keywords: public-key encryption · error-erasure decoding · code-based cryptography

1 Introduction

The McEliece public-key encryption (PKE) scheme is considered the leading code-based candidate in the NIST competition for standardizing post-quantum cryptography. However, it suffers from very large key sizes and relies on the so-called Goppa distinguishing assumption, which has recently come under increasingly severe attacks [16]. Hence, various alternatives have been proposed in the last decades.

The line of cryptosystems we will be interested in began their history in 2003 with the Augot and Finiasz [2,3] cryptosystem. They described a PKE

using Reed-Solomon (RS) codes, where the security relies on a problem known as polynomial reconstruction. In contrast to the McEliece system, the scheme does not rely on hiding a code and it has a simple description where the public key and ciphertext could be seen as corrupted codewords of an RS code. Although the scheme required large field sizes, it provided much smaller keys than McEliece. In 2006, Faure and Loidreau [7] proposed a rank-metric version using Gabidulin codes, where the security was related to a linearized polynomial version of the polynomial reconstruction problem. Since generic decoding attacks in the rank metric are known to be much more expensive than in the Hamming metric, their proposed parameters were even smaller than those of Augot-Finiasz. However, the Hamming and rank-metric schemes were eventually broken by Coron [6] and Gaborit, Otmani, and Talé-Kalachi [9], respectively. The history of these schemes concludes with two recent repairs of the Faure-Loidreau system: RAMESSES [12] and LIGA [17]. Unfortunately, these attempts did not succeed, as was shown by Bombar and Couvreur [5].

In the McEliece scheme, the ciphertext is a codeword plus an error, and the secret is some hidden property of the code which allows for efficient decoding. In this scheme, the ciphertext is a codeword plus an error, but the secret is some property *of the error vector* rather than the code. Hence, the code itself can be made public. Even though the scheme was broken when used with RS or Gabidulin codes, the general idea of this scheme is promising and somewhat natural. Therefore, an important question is whether this idea can be realized using a different code family.

1.1 Contributions

We introduce a PKE framework that generalizes the Augot-Finiasz and the Faure-Loidreau system to new code families. The correctness and security of this framework are equivalent to simple assertions about this family. Specifically, security reduces to a hard problem called *supercode decoding*, and correctness is equivalent to the existence of an efficient error-erasure decoder for this code class. We give a largely unified description of the Augot-Finiasz scheme and its successors, along with how they were attacked. Further, our reduction allows us to show that various other code families would be insecure as well. A question left open is whether there exists a secure choice for the code family or, conversely, if the framework (as stated) is uninstantiable. Even so, we think the general idea of this framework is promising and it is a somewhat natural counterpart to the McEliece paradigm. Additionally, while the framework as stated might be insecure, various modifications are open for further study (for example, using a different “randomization” of the public key in the encryption step), that might eventually lead to secure instantiations.

1.2 Organization of the Paper

Section 2 provides the required background and notation. We describe the framework in Section 3 and state the conditions for its correctness and security in

Section 4. The previously proposed schemes this framework was inspired from are surveyed in Section 5. Finally, Section 6 describes which code classes are insecure and concludes the paper.

2 Preliminaries

Notation. The finite field of size q^m is denoted by $\mathbb{F}_{q^m}$ where q is a prime power and m is a nonzero integer. The notation $\mathbb{F}_{q^m}^*$ refers to the set of its nonzero elements. We write $\alpha \xleftarrow{\$} \mathcal{S}$ to mean that α is drawn uniformly at random from the set $\mathcal{S}$.

For a matrix $\mathbf{X} \in \mathbb{F}_{q^m}^{k \times n}$ and a vector $\mathbf{x} \in \mathbb{F}_{q^m}^n$, the rank of $\mathbf{X}$ over $\mathbb{F}_{q^m}$ is denoted by $\text{rank}(\mathbf{X})$ and the rank of the $\mathbb{F}_q$ -space spanned by the entries $x_1, \dots, x_n$ of $\mathbf{x}$ is written $\text{rank}(\mathbf{x})$. For clarity, we will write $\text{rank}_{\mathbb{F}_{q^m}}(\mathbf{X})$ and $\text{rank}_{\mathbb{F}_q}(\mathbf{x})$. We denote by $\langle \mathbf{X} \rangle$ the $\mathbb{F}_{q^m}$ -subspace spanned by the rows of $\mathbf{X}$ and by $\langle \mathbf{x} \rangle$ the $\mathbb{F}_q$ -space spanned by the entries of $\mathbf{x}$. To avoid ambiguity, these are sometimes written as $\langle \mathbf{X} \rangle_{\mathbb{F}_{q^m}}$ and $\langle \mathbf{x} \rangle_{\mathbb{F}_q}$. Likewise, for a list of vectors $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)} \in \mathbb{F}_{q^m}^n$, their $\mathbb{F}_{q^m}$ -span is $\langle \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)} \rangle_{\mathbb{F}_{q^m}}$. If $\mathcal{U}$ is a subspace over $\mathbb{F}_q$, then $|\mathcal{U}|$ refers to its dimension, and if $\mathcal{U}$ is just a set, then the same notation denotes its cardinality.

We denote by $\mathcal{P}$, respectively $\mathcal{L}$, the set of polynomials, respectively linearized polynomials [14], in the variable x and coefficients in $\mathbb{F}_{q^m}$. Further, notation like $\mathcal{P}_{<k}$ ($\mathcal{L}_{<k}$) will denote (linearized) polynomials with (q -)degree less than k . For any (q -)polynomial f and a vector $\mathbf{x} \in \mathbb{F}_{q^m}^n$, we denote by $f(\mathbf{x})$ its element-wise evaluation, i.e., $(f(x_1), \dots, f(x_n)) \in \mathbb{F}_{q^m}^n$.

Finally, $\mathbf{x}^q$ denotes component-wise exponentiation, i.e., $(x_1^q, \dots, x_n^q) \in \mathbb{F}_{q^m}^n$ and, if $\mathbf{y} \in \mathbb{F}_{q^m}^n$, then $\mathbf{x} \cdot \mathbf{y} = (x_1 y_1, \dots, x_n y_n)$ is the component-wise product of the two vectors.

Hamming Metric. The Hamming weight of a vector in $\mathbb{F}_{q^m}^n$ is the number of its nonzero entries. The *support* of a vector $\mathbf{x} \in \mathbb{F}_{q^m}^n$ is denoted by $\text{supp}(\mathbf{x})$ and is the set of indices of its nonzero entries. In other words,

$$\text{supp}(\mathbf{x}) = \{i \in \{1, \dots, n\} \mid x_i \neq 0\}.$$

Rank Metric. The rank weight of a vector $\mathbf{x} \in \mathbb{F}_{q^m}^n$ over $\mathbb{F}_q$ is given by the dimension of the $\mathbb{F}_q$ -subspace spanned by its entries. Let $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ be a basis of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$. Then we can define the expansion map

$$\begin{aligned} \Gamma : \mathbb{F}_{q^m} &\rightarrow \mathbb{F}_q^m, \\ \sum_{i=1}^m \gamma_i x_i &\mapsto (x_1, \dots, x_m). \end{aligned}$$

By extending this map component-wise to vectors, i.e., $\Gamma : \mathbb{F}_{q^m}^n \rightarrow \mathbb{F}_q^{m \times n}$, we can equivalently define the rank weight of $\mathbf{x}$ as the rank of $\Gamma(\mathbf{x})$. This gives rise

to two types of support: the row support is the row span of $\Gamma(\mathbf{x})$ which is a subspace in $\mathbb{F}_q^n$ and the column support is the column span of $\Gamma(\mathbf{x})$ which is in $\mathbb{F}_q^m$. In the following, $\text{supp}(\mathbf{x})$ will denote the row support unless specified otherwise.

Erasure Decoding. Given a code $\mathcal{C}$, the *error decoding* problem is to recover $\mathbf{c} \in \mathcal{C}$ given the received word $\mathbf{r} = \mathbf{c} + \mathbf{e}$ corrupted by an unknown error vector $\mathbf{e}$ of weight w . This becomes an *erasure decoding* problem if we also know the support of this error. In the Hamming metric, this means we know the location of the nonzero entries of $\mathbf{e}$ but not its values. In the rank metric, this means we know the row (or column) support of $\mathbf{e}$.

Error-Erasure Decoding. Suppose $\mathbf{r} = \mathbf{c} + \mathbf{e}' + \mathbf{e}$ where we have two unknown errors $\mathbf{e}'$ and $\mathbf{e}$ of weights w' and w , respectively. If we are also given the support of $\mathbf{e}'$ then this becomes an error-erasure decoding problem with erasure weight w' and error weight w . Many codes have efficient error-erasure decoders; for instance, Reed-Solomon codes and Gabidulin codes can correct w' erasures and w errors whenever $w' + 2w < d$, where d is the minimum distance of the code.

3 The PKE Framework

We keep the metric unspecified, as one can use either the Hamming or rank metric.

Algorithm. Fix a generator matrix $\mathbf{G} \in \mathbb{F}_{q^m}^{k \times n}$ of a code $\mathcal{C}$ that has an efficient error-erasure decoder $\text{error-erasure-dec}(\cdot)$. Let w' and w be large and small integers, respectively. This code, its decoder, and w' and w are public knowledge.

Key Generation (Algorithm 1) Alice draws random vectors $\mathbf{m}' \in \mathbb{F}_{q^m}^k$ and $\mathbf{e}' \in \mathbb{F}_{q^m}^n$ such that $\text{wt}(\mathbf{e}') = w'$. The public key is $\boldsymbol{\rho} = \mathbf{m}'\mathbf{G} + \mathbf{e}'$, i.e., a random codeword of $\mathcal{C}$ corrupted by some large error. The secret key is $\boldsymbol{\kappa} = \mathbf{e}'$.

Encryption (Algorithm 2) Given a message $\mathbf{m} \in \mathbb{F}_{q^m}^k$, Bob picks a random $\alpha \in \mathbb{F}_{q^m}^*$ and a random vector $\mathbf{e} \in \mathbb{F}_{q^m}^n$ of weight w . The ciphertext is given by $\mathbf{r} = \mathbf{m}\mathbf{G} + \alpha\boldsymbol{\rho} + \mathbf{e}$.

Decryption (Algorithm 3) Note that the ciphertext $\mathbf{r}$ is a corrupted codeword of $\mathcal{C}$ with error $\alpha\mathbf{e}' + \mathbf{e}$. Since multiplication with the scalar α preserves the support of $\mathbf{e}'$ ³, Alice knows $\text{supp}(\mathbf{e}') = \text{supp}(\alpha\mathbf{e}')$ and can perform error-erasure decoding to obtain $\alpha\mathbf{e}' + \mathbf{e}$. With $\alpha\mathbf{e}' + \mathbf{e}$ in hand, it is possible to extract α and hence find $\mathbf{m}$. Concrete examples of this extraction step in the Hamming and rank metric are given after the next paragraph.

³ Note this holds for the Hamming metric and, assuming we use the concept of row support, also for the rank metric.

Algorithm 1: Keygen

Input: —
Output: public key $\rho \in \mathbb{F}_{q^m}^n$,
secret key $\kappa \in \mathbb{F}_{q^m}^n$

- 1 $e' \xleftarrow{\$} \mathbb{F}_{q^m}^n$ s.t. $\text{wt}(e') = w'$
- 2 $m' \xleftarrow{\$} \mathbb{F}_{q^m}^k$
- 3 $\rho \leftarrow m'G + e'$
- 4 $\kappa \leftarrow e'$
- 5 **return** (ρ, κ)

Algorithm 2: Encryption

Input: message $m \in \mathbb{F}_{q^m}^k$,
Public key $\rho \in \mathbb{F}_{q^m}^n$
Output: ciphertext $r \in \mathbb{F}_{q^m}^n$

- 1 $\alpha \xleftarrow{\$} \mathbb{F}_{q^m}^*$
- 2 $e \xleftarrow{\$} \mathbb{F}_{q^m}^n$ s.t. $\text{wt}(e) = w$
- 3 $r \leftarrow mG + \alpha\rho + e$
- 4 **return** r

Algorithm 3: Decryption

Input: ciphertext $r \in \mathbb{F}_{q^m}^n$, secret key $\kappa = e' \in \mathbb{F}_{q^m}^n$
Output: message $\hat{m} \in \mathbb{F}_{q^m}^k$

- 1 $x \leftarrow \text{error-erasure-dec}(r, \text{supp}(e'))$ // x should be $\alpha e' + e$
- 2 $\hat{\alpha} \leftarrow \text{extract } \alpha \text{ from } x$ // see examples below
- 3 Solve $\hat{m}G = r - x - \hat{\alpha}m'G$ for $\hat{m}$ // $r - x$ should equal $(m + \alpha m')G$
- 4 **return** $\hat{m}$

Intuition. The code $\mathcal{C} = \langle G \rangle$ has a publicly known decoder that can efficiently correct a certain number of errors and erasures. The public key ρ is a corrupted codeword of $\mathcal{C}$ with an error weight w' far above the decoder's error correction capability. The idea for encryption is as follows: Bob wants to add an error to mG such that Alice can decode it but not anyone else. Since the public key ρ contains a large error of weight w' that is only known to Alice, one possibility is for Bob to add a random multiple of ρ to obtain the ciphertext $\tilde{r} = mG + \alpha\rho$. The error added is $\alpha e'$ and its support is known only to Alice. Hence, for Alice, $\tilde{r}$ contains only erasures while for an attacker $\tilde{r}$ contains errors. However, this would not be secure as the ciphertext is just a codeword of the known extended code $\left\langle \begin{bmatrix} G \\ \rho \end{bmatrix} \right\rangle$ and thus $\begin{bmatrix} m \\ \alpha \end{bmatrix}$ can be obtained by matrix inversion. To prevent this attack, Bob also adds a small weight- w error e of his own. Hence, the final ciphertext expression is $r = mG + \alpha\rho + e$. Thus, if Alice can decode w' erasures and w errors in this code, then she can correctly decrypt. However, an attacker seemingly faces the problem of decoding at least w' errors. With a suitable code and parameter choice, one expects the former problem to be easy and the latter to be hard.

Example: Hamming Metric. Let us give more details for the decryption step.

The first step consists of error-erasure decoding the ciphertext $\mathbf{r}$ to obtain $\alpha\mathbf{e}' + \mathbf{e}$, while the second step extracts α from $\alpha\mathbf{e}' + \mathbf{e}$. After this, Alice can easily find $\mathbf{m}$.

- We illustrate error-erasure decoding by assuming the code is an RS code. Since

$$\mathbf{r} = \mathbf{m}\mathbf{G} + \alpha\mathbf{p} + \mathbf{e} = (\mathbf{m} + \alpha\mathbf{m}')\mathbf{G} + (\alpha\mathbf{e}' + \mathbf{e}),$$

if Alice punctures $\mathbf{r}$ at $\text{supp}(\mathbf{e}')$, then the $\alpha\mathbf{e}'$ term disappears and she is left with a corrupted codeword of error weight at most w in a punctured RS code of blocklength $n - w'$, which is still an RS code. Thus, if $w \leq \frac{n-w'-k}{2}$, she can decode to obtain $\mathbf{m} + \alpha\mathbf{m}'$ or equivalently $\alpha\mathbf{e}' + \mathbf{e}$.

- Since $w < w'$, several entries of $\alpha\mathbf{e}' + \mathbf{e}$ must be equal to $\alpha e'_i$ and hence reveal α .
- Since Alice knows $\mathbf{m}'$, α and $(\mathbf{m} + \alpha\mathbf{m}')$, she can find $\mathbf{m}$ (step 3 of Algorithm 3).

Example: Rank Metric. We again explain the decryption in more detail, where the first two steps consist of error-erasure decoding and extraction.

- We illustrate error-erasure decoding assuming a Gabidulin code is used. Since Alice knows the row support of $\mathbf{e}'$, she knows a matrix $\mathbf{J} \in \mathbb{F}_q^{n \times n}$ such that the vector $\mathbf{e}'\mathbf{J} \in \mathbb{F}_q^n$ is zero in the first $n - w'$ positions. Note that

$$\mathbf{r}\mathbf{J} = \mathbf{m}\mathbf{G}\mathbf{J} + \alpha\mathbf{p}\mathbf{J} + \mathbf{e}\mathbf{J} = (\mathbf{m} + \alpha\mathbf{m}')\mathbf{G}\mathbf{J} + (\alpha\mathbf{e}'\mathbf{J} + \mathbf{e}\mathbf{J}).$$

Thus, if we puncture in the last w' positions, the $\alpha\mathbf{e}'\mathbf{J}$ term disappears and we are left with an error-only decoding problem with error weight equal to $\text{rank}_{\mathbb{F}_q}(\mathbf{e}\mathbf{J}) \leq w$. This decoding problem is in a code of blocklength $n - w'$, obtained by puncturing the last w' positions of $\langle \mathbf{G}\mathbf{J} \rangle$. This is still a Gabidulin code and, if $w \leq \frac{n-w'-k}{2}$, Alice can decode to obtain $\alpha\mathbf{e}' + \mathbf{e}$.

- Extracting α can be done as follows. The error-erasure decoder outputs $\mathbf{x} = \alpha\mathbf{e}' + \mathbf{e}$. Multiplying this with the same $\mathbf{J}$ as above, Alice gets $\mathbf{x}\mathbf{J} = \mathbf{e}'\mathbf{J} + \mathbf{e}\mathbf{J}$. Note the first $n - w'$ positions of $\mathbf{x}\mathbf{J}$ equal $\mathbf{e}\mathbf{J}$ and with high probability the $\mathbb{F}_q$ -span of these $n - w'$ positions will equal $\langle \mathbf{e} \rangle_{\mathbb{F}_q}$. Alice picks any $\mathbf{a} \in \mathbb{F}_q^w$ such that $\langle \mathbf{a} \rangle_{\mathbb{F}_q} = \langle \mathbf{e} \rangle_{\mathbb{F}_q}$. She can now write $\mathbf{x} = \alpha\mathbf{e}' + \mathbf{a}\mathbf{B}$ where only $\alpha \in \mathbb{F}_q^*$ and $\mathbf{B} \in \mathbb{F}_q^{w \times n}$ are unknown. This is a linear system of mn equations over $\mathbb{F}_q$ with $m + nw$ unknowns. Since w is small, this system is overdetermined and hence has a unique solution with high probability.
- As before, since Alice knows $\mathbf{m}'$, α and $(\mathbf{m} + \alpha\mathbf{m}')$, she can find $\mathbf{m}$.

The framework can also be used in the rank metric using the concept of column support instead. In this case, Bob needs to multiply $\mathbf{p}$ by an α drawn from $\mathbb{F}_q^*$ instead of $\mathbb{F}_{q^m}^*$, in order to keep the column support unchanged. Then Alice can decrypt assuming she has an error-erasure decoder, with erasures here referring to knowing the column support.

4 Security and Correctness

The following assertions are easy to show.

Correctness The framework is correct if the code family has an error-erasure decoder that can efficiently correct w errors and w' erasures. In the Hamming metric, decryption is always successful, while in the rank metric, we tolerate a small decryption failure probability since the extraction step can fail sometimes.

Security The framework has one-wayness under chosen plaintext attack (OW-CPA) security if the following problem is hard. Let $\mathcal{S}_w$ denote the set of vectors in $\mathbb{F}_{q^m}^n$ of weight w . The problem is parameterized by a public generator matrix $\mathbf{G} \in \mathbb{F}_{q^m}^{k \times n}$ and integers w', w satisfying $w' > w$.

Problem 1 (Supercode Decoding $(\mathbf{G}, w, w')$). Draw $\mathbf{c}' \in \langle \mathbf{G} \rangle$ and $\mathbf{e}' \in \mathcal{S}_{w'}$ uniformly at random. Set $\boldsymbol{\rho} = \mathbf{c}' + \mathbf{e}'$. Draw $\mathbf{c}^\dagger \in \left\langle \begin{bmatrix} \mathbf{G} \\ \boldsymbol{\rho} \end{bmatrix} \right\rangle \setminus \langle \mathbf{G} \rangle$ and $\mathbf{e} \in \mathcal{S}_w$ uniformly at random. Set $\mathbf{r} = \mathbf{c}^\dagger + \mathbf{e}$. Given only $\mathbf{G}$, $\boldsymbol{\rho}$ and $\mathbf{r}$, find $\mathbf{e}$.

An open question, then, is whether there exists a code and integers w', w such that the above assertions hold simultaneously.

5 A Survey of Past Schemes

Our framework is inspired by and generalizes the Augot-Finiasz and Faure-Loidreau systems. However, this generalization is not exact and these proposals exhibit some differences. To better understand these past schemes, we introduce a *shorter message* and an *interleaved* variation of the framework.

5.1 Shorter Message Variation

In this variation, the first entry of the message $\mathbf{m}$ is required to be 0 while the first entry of $\mathbf{m}'$ is nonzero. This simplifies the decryption algorithm: after the error-erasure decoder returns $\alpha \mathbf{e}' + \mathbf{e}$, or equivalently $\alpha \mathbf{m}' + \mathbf{m}$, we can recover α directly by comparing the first entry of $\alpha \mathbf{m}' + \mathbf{m}$ with the first entry of $\mathbf{m}'$. In specific cases, the shorter message variation has benefits. In the case of rank metric, there is no longer any decryption failure probability, and for the interleaved variant in the Hamming metric, this shorter message technique is even required to make decryption work. The big drawback, in our view, is that the underlying hard problem becomes more ad hoc and complex.

5.2 Interleaved Variation

Key Generation (Algorithm 4) In this variant, the public key consists of ℓ corrupted codewords, where ℓ is called the interleaving order. That is, $\boldsymbol{\rho} =$

Algorithm 4: Keygen (Interleaving)

Input: —
Output: public key $\rho \in (\mathbb{F}_{q^m}^n)^\ell$, secret key $\kappa \in (\mathbb{F}_{q^m}^n)^\ell$

- 1 $e^{(1)}, \dots, e^{(\ell)} \xleftarrow{\$} \mathbb{F}_{q^m}^n$
s.t. $|\cup_{i=1}^\ell \text{supp}(e^{(i)})| = w'$ if Hamming metric
or $|\text{supp}(e^{(1)}) + \dots + \text{supp}(e^{(\ell)})| = w'$ if rank metric
- 2 $m^{(1)}, \dots, m^{(\ell)} \xleftarrow{\$} \mathbb{F}_{q^m}^k$
- 3 $\rho \leftarrow (\rho^{(1)}, \dots, \rho^{(\ell)})$ where $\rho^{(i)} \leftarrow m^{(i)}\mathbf{G} + e^{(i)}$
- 4 $\kappa \leftarrow (e^{(1)}, \dots, e^{(\ell)})$
- 5 **return** (ρ, κ)

Algorithm 5: Encryption (Interleaving)

Input: message $m \in \mathbb{F}_{q^m}^k$, public key $\rho \in (\mathbb{F}_{q^m}^n)^\ell$
Output: ciphertext $r \in \mathbb{F}_{q^m}^n$

- 1 $\alpha_1, \dots, \alpha_\ell \xleftarrow{\$} \mathbb{F}_{q^m}$ not all zero
- 2 $e \xleftarrow{\$} \mathbb{F}_{q^m}^n$ s.t. $\text{wt}(e) = w$
- 3 $r \leftarrow m\mathbf{G} + \sum_{i=1}^\ell \alpha_i \rho^{(i)} + e$
- 4 **return** r

$(\rho^{(1)}, \dots, \rho^{(\ell)})$ where $\rho^{(i)} = m^{(i)}\mathbf{G} + e^{(i)}$ for $i = 1, \dots, \ell$. We choose $m^{(i)}$ independently and uniformly at random. The error vectors $e^{(i)}$ are random weight- w' vectors but with the additional constraint that their support is confined to a common support.⁴ In the Hamming metric, this means $|\cup_{i=1}^\ell \text{supp}(e^{(i)})| = w'$ while the condition in the rank metric becomes $|\text{supp}(e^{(1)}) + \dots + \text{supp}(e^{(\ell)})| = w'$.⁵

Encryption (Algorithm 5) Bob chooses random $\alpha_i \in \mathbb{F}_{q^m}$, not all zero, for $i = 1, \dots, \ell$ and an error vector e of weight w . The ciphertext is then given by $r = m\mathbf{G} + \sum_{i=1}^\ell \alpha_i \rho^{(i)} + e$. That is, Bob adds a random non-zero linear combination of the $\rho^{(i)}$'s and a small weight error vector.

Decryption (Algorithm 6) Let us first consider the rank metric. Alice knows a w' -dimensional space containing the support of all the errors $e^{(i)}$ and thus, in particular, the support of $\sum_{i=1}^\ell \alpha_i e^{(i)}$. Hence, she can obtain $e + \sum_i \alpha_i e^{(i)}$ by error-erasure decoding r . Then, using an extraction step similar to the non-interleaved case, she can find $\alpha_1, \dots, \alpha_\ell$ with high probability and hence m .

⁴ Actually, the key generation could be modified so that the $e^{(i)}$'s are also linearly independent over $\mathbb{F}_{q^m}$, since this is necessary for the extraction step during decryption. But, as we will see later, this is only relevant in the rank metric since extraction is inefficient in the Hamming metric anyway.

⁵ Recall that, for the rank metric, $\text{supp}(e^{(i)}) \subseteq \mathbb{F}_q^n$ is the row support of e' and $|\mathcal{U}|$ denotes the dimension of the subspace $\mathcal{U}$.

Algorithm 6: Decryption (Interleaving)

Input: ciphertext $\mathbf{r} \in \mathbb{F}_{q^m}^n$, secret key $\boldsymbol{\kappa} = (\mathbf{e}'^{(1)}, \dots, \mathbf{e}'^{(\ell)}) \in (\mathbb{F}_{q^m}^n)^\ell$
Output: message $\hat{\mathbf{m}} \in \mathbb{F}_{q^m}^k$

- 1 $\mathbf{x} \leftarrow \text{error-erasure-dec}(\mathbf{r}, \cup_{i=1}^\ell \text{supp}(\mathbf{e}'^{(i)}))$ if Hamming metric
or $\mathbf{x} \leftarrow \text{error-erasure-dec}(\mathbf{r}, \text{supp}(\mathbf{e}'^{(1)}) + \dots + \text{supp}(\mathbf{e}'^{(\ell)}))$ if rank metric
// $\mathbf{x}$ should be $\sum_{i=1}^\ell \alpha_i \mathbf{e}'^{(i)} + \mathbf{e}$
- 2 $\hat{\alpha}_1, \dots, \hat{\alpha}_\ell \leftarrow \text{extract } \alpha_1, \dots, \alpha_\ell \text{ from } \mathbf{x}$
// inefficient in Hamming metric
- 3 $\hat{\mathbf{m}} \leftarrow \text{solve } \hat{\mathbf{m}}\mathbf{G} = \mathbf{r} - \mathbf{x} - \sum_{i=1}^\ell \hat{\alpha}_i \mathbf{m}'^{(i)}\mathbf{G} \text{ for } \hat{\mathbf{m}}$
// $\mathbf{r} - \mathbf{x}$ should equal $(\mathbf{m} + \sum_{i=1}^\ell \alpha_i \mathbf{m}'^{(i)})\mathbf{G}$
- 4 **return** $\hat{\mathbf{m}}$

The Hamming metric case is similar: Alice knows a w' -size set of indices containing the support of all the errors $\mathbf{e}'^{(i)}$ and thus, in particular, the support of $\sum_{i=1}^\ell \alpha_i \mathbf{e}'^{(i)}$. Hence, she can obtain $\mathbf{e} + \sum_i \alpha_i \mathbf{e}'^{(i)}$ by error-erasure decoding $\mathbf{r}$. However, there does not seem to be an efficient way to extract $\alpha_1, \dots, \alpha_\ell$ from $\mathbf{e} + \sum_i \alpha_i \mathbf{e}'^{(i)}$ unless we use the shorter message variation described next.

Shorter Message Variation with Interleaving With this variation, the first ℓ entries of $\mathbf{m}$ are required to be 0 and the first ℓ entries of each $\mathbf{m}'^{(i)}$ are set to zero except the i th entry which is set to 1.⁶ Thus, the first ℓ entries of $\mathbf{m} + \sum_{i=1}^\ell \alpha_i \mathbf{m}'^{(i)}$ equal $(\alpha_1, \dots, \alpha_\ell)$ and hence finding $\mathbf{m}$ is straightforward after the error-erasure decoder returns $\mathbf{m} + \sum_{i=1}^\ell \alpha_i \mathbf{m}'^{(i)}$.

Intuitively, the interleaved version might be harder to break since there are more unknowns α_i in the ciphertext. However, as in the shorter message variation, the big drawback is that the system and the underlying hard problem (Appendix A) becomes more complex to analyse.

5.3 Framework with RS and Gabidulin Codes

We describe our framework instantiated with RS and Gabidulin codes and explain why these choices are insecure. Many of the previously proposed schemes and the attacks against them can be seen in this framework. Since several of these schemes also use the shorter message and/or the interleaved variation, we also account for them.

Remark 1. The length of the message vector $\mathbf{m}$ is k , but for the past schemes that use the shorter message variation, the first ℓ entries will be set to zero (where ℓ denotes the interleaving order and $\ell = 1$ in the non-interleaved case). We will denote this “effective length” of $\mathbf{m}$ with k' . We have $k' = k - \ell$ if we use the shorter message variation, else $k' = k$.

⁶ This can be generalized to the requirement that the vectors $\boldsymbol{\rho}_{1:\ell}^{(1)}, \boldsymbol{\rho}_{1:\ell}^{(2)}, \dots, \boldsymbol{\rho}_{1:\ell}^{(\ell)}$ are linearly independent where $\mathbf{x}_{1:\ell}$ denotes the first ℓ entries of the vector $\mathbf{x}$.

Assume that we instantiate the scheme in the Hamming metric using an RS code with evaluation points $\mathbf{a} = (a_1, \dots, a_n)$. Recall that the public key and the ciphertext are given by

$$\boldsymbol{\rho} = \mathbf{m}'\mathbf{G} + \mathbf{e}', \quad (1)$$

$$\mathbf{r} = \mathbf{m}\mathbf{G} + \alpha\boldsymbol{\rho} + \mathbf{e}. \quad (2)$$

The linearization attack due to Coron [6] works as follows:⁷ let $V_0 \in \mathcal{P}_{\leq w}$ be a polynomial of degree w such that for all i we have $V_0(a_i) \cdot e_i = 0$, i.e., $V_0(\mathbf{a}) \cdot \mathbf{e} = \mathbf{0}$. Then, if we component-wise multiply (2) by $V_0(\mathbf{a})$, we get

$$V_0(\mathbf{a}) \cdot \mathbf{r} = V_0(\mathbf{a}) \cdot \mathbf{m}(\mathbf{a}) + V_0(\mathbf{a}) \cdot \alpha\boldsymbol{\rho} \quad (3)$$

where $m \in \mathcal{P}_{< k'}$ is the polynomial such that $m(\mathbf{a}) = \mathbf{m}\mathbf{G}$. We now linearize this system of equations. Let $V \in \mathcal{P}_{\leq w}$, $N \in \mathcal{P}_{< w+k'}$ and $R \in \mathcal{P}_{\leq w}$ be polynomials such that

$$V(\mathbf{a}) \cdot \mathbf{r} = N(\mathbf{a}) + R(\mathbf{a}) \cdot \boldsymbol{\rho}. \quad (4)$$

The first set of equations (3) has a non-zero solution (V_0, m, α) by the problem statement. Since the second set of equations (4) is more general than the first (3), this non-zero solution also yields a solution $(V = V_0, N = V_0 \cdot m, R = \alpha V_0)$ of (4). The second system of equations is homogeneous and linear, and since it has a non-zero solution, its kernel has dimension at least one. If we assume this dimension to be exactly one, then the solution space is generated by (V_0, m, α) and so (3) can be solved by solving (4). The latter has $(w+1) + (w+k') + (w+1) = k' + 3w + 2$ unknowns and n equations. If $n \geq k' + 3w + 1$, then the kernel has with high probability dimension one, as observed experimentally in [6]. Thus, the system can be broken.

Now we consider the generalization of this attack to the interleaved variant. Here the public key and ciphertext expressions are

$$\boldsymbol{\rho}^{(i)} = \mathbf{m}'^{(i)}\mathbf{G} + \mathbf{e}'^{(i)} \quad \text{for } i = 1, \dots, \ell, \quad (5)$$

$$\mathbf{r} = \mathbf{m}\mathbf{G} + \sum_{i=1}^{\ell} \alpha_i \boldsymbol{\rho}^{(i)} + \mathbf{e}. \quad (6)$$

Then, as before, we can set up the following two systems of equations. Let $V_0 \in \mathcal{P}_{\leq w}$ be such that

$$V_0(\mathbf{a}) \cdot \mathbf{r} = V_0(\mathbf{a}) \cdot \mathbf{m}(\mathbf{a}) + V_0(\mathbf{a}) \cdot \sum_{j=1}^{\ell} \alpha_j \boldsymbol{\rho}^{(j)}. \quad (7)$$

⁷ In this non-interleaved case, Coron actually derived an improved attack using determinants that did not involve linearizing the αV term. This had the advantage that his attack was not probabilistic and had a more relaxed requirement on the parameters for the attack to work.

Let $V \in \mathcal{P}_{\leq w}$, $N \in \mathcal{P}_{< w+k'}$ and $R_1, \dots, R_\ell \in \mathcal{P}_{\leq w}$ be such that

$$V(\mathbf{a}) \cdot \mathbf{r} = N(\mathbf{a}) + \sum_{j=1}^{\ell} R_\ell(\mathbf{a}) \cdot \boldsymbol{\rho}^{(j)}. \quad (8)$$

The latter system has $(w+1) + (w+k') + \ell(w+1) = (\ell+2)w + k' + \ell + 1$ unknowns and n equations. As before, this system will have a kernel of dimension one with high probability if $n \geq (\ell+2)w + k' + \ell$.

The attack in the rank metric using Gabidulin codes is similar, where linearized polynomials are used instead. Let $\mathbf{g} = (g_1, \dots, g_n) \in \mathbb{F}_{q^m}^n$ be the evaluation points of the Gabidulin code, where $g_1, \dots, g_n$ are linearly independent over $\mathbb{F}_q$. Let $V_0 \in \mathcal{L}_{\leq w}$ be a q -polynomial of q -degree w such that for all i we have $V_0(e_i) = 0$, i.e., $V_0(\mathbf{e}) = \mathbf{0}$. Such a V_0 exists since, for any given $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^m}$ of dimension w , it is always possible to find a q -polynomial of q -degree w whose kernel equals this subspace [14, Theorem 7]. We have

$$V_0(\mathbf{r}) = V_0 \circ m(\mathbf{g}) + \sum_{j=1}^{\ell} V_0(\alpha_j \boldsymbol{\rho}^{(j)}) \quad (9)$$

where $m \in \mathcal{L}_{< k'}$ is the q -polynomial such that $m(\mathbf{g})$ equals $\mathbf{mG}$. Let $V \in \mathcal{L}_{\leq w}$, $N \in \mathcal{L}_{< w+k'}$ and $R \in \mathcal{L}_{\leq w}$ be such that

$$V(\mathbf{r}) = N(\mathbf{g}) + \sum_{j=1}^{\ell} R_\ell(\boldsymbol{\rho}^{(j)}). \quad (10)$$

For a solution of the second system to yield a solution to the first, we again require $n \geq (\ell+2)w + k' + \ell$.

Is it possible to choose parameters such that $n < (\ell+2)w + k' + \ell$ in order to thwart this attack? The obvious solution of arbitrarily increasing ℓ does not work. It is well-known that interleaved Reed-Solomon codes (see, e.g., [4, 18]) and interleaved Gabidulin codes (see [13, 19] for common column support and [15, Section 3.2] for common row support) can be decoded with high probability up to $\frac{\ell}{\ell+1}(n-k)$ errors, where the number of “errors” here refers to the size/dimension of the common support shared by these ℓ error vectors. Thus, if $w' \leq \frac{\ell}{\ell+1}(n-k)$ then it becomes possible to recover the private key $(\mathbf{e}'^{(1)}, \dots, \mathbf{e}'^{(\ell)})$ from the public key $(\boldsymbol{\rho}^{(1)}, \dots, \boldsymbol{\rho}^{(\ell)})$. As a result, we have the following conditions on the cryptosystem:

$$2w + w' \leq n - k \quad \text{for correct decryption,} \quad (11)$$

$$n < (\ell+2)w + k' + \ell \quad \text{to prevent linearization attack,} \quad (12)$$

$$w' > \frac{\ell}{\ell+1}(n-k) \quad \text{to prevent interleaved decoding attack.} \quad (13)$$

It can be shown that these inequalities are incompatible, see Appendix B, and thus the system is insecure with both RS and Gabidulin codes.

5.4 Past Schemes

Augot-Finiasz Scheme (2003) [2,3]. This scheme is obtained by instantiating our framework with RS codes in the Hamming metric. The only difference is that Augot and Finiasz use the shorter message variant (which, in this case, provides no advantage in our view). In a subsequent work [3], they introduce the interleaved version of their scheme, which they do so in terms of the trace operator. Both were successfully attacked by Coron’s linearization attack [6] in 2004, which we explained in Section 5.3. Coron also showed that increasing the interleaving order ℓ cannot help by describing an algorithm, which is equivalent to an interleaved RS decoder, that recovers the private key. Kiayias and Yung [11] showed that the non-interleaved version of the system is still insecure if we increase w by (a) using Guruswami-Sudan list decoding in the decryption step instead of unique decoding and (b) using the fact that some of the w errors will overlap with the w' errors in the public key and thus a larger w can still allow decryption (with a small probability of failure).

Faure-Loidreau Scheme (2006) [7] This is the rank-metric counterpart of the Augout-Finiasz scheme using Gabidulin codes. The paper noted that the non-interleaved version is vulnerable to a Coron-like linearization attack and put forward the interleaved variant with a large ℓ to avoid this attack. As the discussion in Section 5.3 shows, if the interleaving order ℓ is chosen large enough to resist the linearization attack, it becomes vulnerable to an interleaved decoding attack. Thus, this scheme was also broken, and this vulnerability was shown by Gaborit et al. [9].

RAMESSES (Lavauzelle et al., 2019) [12] This scheme was proposed as a repair of the non-interleaved variant of the Faure-Loidreau cryptosystem. A new feature of their scheme is that they preserve the column support of the error vector $\mathbf{e}'$ instead of the row support. They formulate the scheme using the parity-check matrix but we will give an equivalent formulation using the generator matrix to keep our presentation unified. We note the scheme sets $n = m$.

Consider the ciphertext expression in the original Faure-Loidreau system,

$$\mathbf{r} = \mathbf{m}\mathbf{G} + \alpha\boldsymbol{\rho} + \mathbf{e}$$

where $\alpha \in \mathbb{F}_{q^m}^*$ and $\mathbf{e} \in \mathbb{F}_{q^m}^n$ of weight w are drawn uniformly at random. To preserve the column support of $\mathbf{e}'$ in $\boldsymbol{\rho}$, we need to draw α from $\mathbb{F}_q^*$ instead of $\mathbb{F}_{q^m}^*$. RAMESSES then goes further and also allows random column operations over $\mathbb{F}_q$ on the vector $\boldsymbol{\rho}$, i.e., right-multiplying $\boldsymbol{\rho}$ by a random invertible matrix $\mathbf{T} \in \mathbb{F}_q^{m \times m}$. Hence the ciphertext becomes

$$\mathbf{r} = \mathbf{m}\mathbf{G} + \boldsymbol{\rho}\mathbf{T} + \mathbf{e}. \quad (14)$$

Notice we do not need α anymore since its effect can be subsumed in $\mathbf{T}$. However, this poses a difficulty for Alice to decrypt. The ciphertext can be written as

$$\mathbf{r} = \mathbf{m}\mathbf{G} + \boldsymbol{\rho}\mathbf{T} + \mathbf{e} = \mathbf{m}\mathbf{G} + \mathbf{m}'\mathbf{G}\mathbf{T} + \mathbf{e}'\mathbf{T} + \mathbf{e}. \quad (15)$$

The $\mathbf{mG}$ term is the codeword, the $\mathbf{e}'\mathbf{T}$ term is an erasure for Alice since she knows its column support (of dimension w') and the $\mathbf{e}$ term is an error (of rank w). But $\mathbf{m}'\mathbf{GT}$ is not necessarily a codeword of $\langle \mathbf{G} \rangle$ and it introduces a potentially large error, thus making $\mathbf{r}$ undecodable for Alice.

To address this, we only use matrices $\mathbf{T}$ such that $\mathbf{gT} \in \langle \mathbf{g}, \mathbf{g}^q, \dots, \mathbf{g}^{q^\gamma} \rangle_{\mathbb{F}_{q^m}}$, where γ is another public parameter of the cryptosystem and $\mathbf{g} \in \mathbb{F}_{q^m}^n$ is the list of evaluation points for the Gabidulin code. Then, for all codewords $\mathbf{c}$ of this $[n, k]$ Gabidulin code, $\mathbf{cT}$ is in the bigger $[n, k + \gamma]$ Gabidulin code defined using the same evaluation points $\mathbf{g}$. In other words, $\mathbf{cT} \in \langle \mathbf{g}, \mathbf{g}^q, \dots, \mathbf{g}^{q^{k+\gamma}} \rangle_{\mathbb{F}_{q^m}}$. This follows because, if $\mathbf{c} = m(\mathbf{g})$ where $m \in \mathcal{L}_{<k}$, then $\mathbf{cT} = m(\mathbf{g})\mathbf{T} = m(\mathbf{gT})$ which is in $\langle \mathbf{g}, \mathbf{g}^q, \dots, \mathbf{g}^{q^{k+\gamma}} \rangle_{\mathbb{F}_{q^m}}$. Alice can then decrypt by error-erasure decoding in this bigger code if

$$2w + w' \leq n - (k + \gamma). \quad (16)$$

The scheme was broken by Bombar and Couvreur [5] where they show how to adapt Coron's linearization attack. Consider the ciphertext expression again:

$$\mathbf{r} = \mathbf{mG} + \rho\mathbf{T} + \mathbf{e}.$$

The attack begins by rewriting this using q -polynomials. Each vector $\mathbf{x} \in \mathbb{F}_{q^m}^m$ corresponds to a unique q -polynomial $f \in \mathcal{L}_{<m}$ such that $\mathbf{x} = (f(g_1), \dots, f(g_m))$, and this can be found using interpolation. Note addition in $\mathbb{F}_{q^m}^m$ corresponds to addition in $\mathcal{L}_{<m}$ under this mapping. Thus, we can write (14) as

$$r = m + j + e$$

where $m \in \mathcal{L}_{<k}$ and $r, e \in \mathcal{L}_{<m}$ are the q -polynomials found by interpolating $\mathbf{mG}, \mathbf{r}$ and $\mathbf{e}$ respectively, and the q -polynomial j is the interpolated version of the vector $\rho\mathbf{T}$. We will now find j . Let $\rho \in \mathcal{L}_{<m}$ be the interpolated version of ρ . Let $T \in \mathcal{L}_{\leq \gamma}$ be the q -polynomial such that $T(\mathbf{g}) = \mathbf{gT}$. The latter is in $\langle \mathbf{g}, \mathbf{g}^q, \dots, \mathbf{g}^{q^\gamma} \rangle_{\mathbb{F}_{q^m}}$ and hence T has q -degree γ .⁸ We have $\rho \circ T(\mathbf{g}) = \rho(\mathbf{gT}) = \rho(\mathbf{g})\mathbf{T} = \rho\mathbf{T}$. Thus $\rho \circ T$ is the interpolated version of $\rho\mathbf{T}$. Hence,

$$r = m + \rho \circ T + e \quad (17)$$

Note that if we left-compose (17) with a $V \in \mathcal{L}_{\leq w}$ such that $V \circ e = 0$ (such a V can be shown to exist) then we get $V \circ r = V \circ m + V \circ \rho \circ T$. Then to linearize $V \circ \rho \circ T$ we would need to replace this entire term with an unknown polynomial of a large degree since we cannot take advantage of the fact that ρ is known as it appears between the unknowns V and T . Thus, this introduces too many unknowns for the attack to work. Indeed, what is needed is a composition from the right, and this is what [5] shows how to do.

⁸ If $\mathbf{T}$ is chosen such that $\mathbf{gT} \in \langle \mathbf{g}, \mathbf{g}^q, \dots, \mathbf{g}^{q^\gamma} \rangle_{\mathbb{F}_{q^m}} \setminus \langle \mathbf{g}, \mathbf{g}^q, \dots, \mathbf{g}^{q^{\gamma-1}} \rangle_{\mathbb{F}_{q^m}}$, then its q -degree is γ otherwise it is at most γ .

Let $V \in \mathcal{L}_{\leq w}$ be a q -polynomial of q -degree at most w such that $e \circ V = 0$. That such a non-zero V exists is proved in [5, Proposition 1]. Thus,

$$r \circ V = m \circ V + \rho \circ T \circ V. \quad (18)$$

We can then linearize $m \circ V$ and $T \circ V$ to get

$$r \circ V = N + \rho \circ R \quad (19)$$

where $N \in \mathcal{L}_{< w+k}$ and $R \in \mathcal{L}_{\leq w+\gamma}$.

The unknowns here are the coefficients of V , N , and R . Note that this is not a linear system since $\rho \circ R$ contains higher degree terms in the coefficients of R . Nonetheless, [5] shows that this equation can be transformed to an equivalent system, which is linear, by using an adjoint operation for q -polynomials. Hence, the attack can be carried out.

Remark 2. The description above is *almost* equivalent to the original RAMESSES scheme, with the following differences. First, our scheme puts the plaintext in $\mathbf{m}$, whereas the original scheme places it in the row support of $\mathbf{e}$. Second, in our scheme, Bob adds the random codeword $\mathbf{m}\mathbf{G}$ which belongs to the $[n, k]$ Gabidulin code. However, the codeword can actually be drawn from the larger $[n, k + \gamma]$ Gabidulin code. With these two changes, our description becomes an equivalent “generator-matrix” version of the original scheme.

LIGA (Renner et al., 2021) [17] This scheme was another repair of the (interleaved variant of the) Faure-Loidreau system. We will first explain the idea of the repair and then show why it turned out to be insecure (shown by [5]). Recall that the public key is given by

$$\boldsymbol{\rho}^{(i)} = \mathbf{m}'^{(i)}\mathbf{G} + \mathbf{e}'^{(i)} \quad \text{where } i = 1, \dots, \ell,$$

where the $\mathbf{e}'^{(i)}$ ’s are such that they share a common row support of dimension w' . That is,

$$\left| \text{supp}(\mathbf{e}'^{(1)}) + \dots + \text{supp}(\mathbf{e}'^{(\ell)}) \right| = w'. \quad (20)$$

As discussed in Section 5.3, if we choose the interleaving order ℓ large enough in order to avoid the linearization attack, the scheme becomes susceptible to an interleaved decoding attack. However, the paper [17] notes that all known interleaved decoders fail if a certain condition [17, Lemma 1] about the ℓ error vectors is satisfied. This latter condition, in turn, is implied by the following [17, Theorem 4],

$$\text{rank}_{\mathbb{F}_{q^m}} \begin{bmatrix} \mathbf{e}'^{(1)} \\ \vdots \\ \mathbf{e}'^{(\ell)} \end{bmatrix} < \frac{w}{n - k - w}. \quad (21)$$

It is possible to pick $\mathbf{e}'^{(i)}$ ’s such that both (20) and (21) hold, and the repair suggests drawing the $\mathbf{e}'^{(i)}$ ’s in the key generation step in precisely this way.

This repair did not succeed because picking linearly dependent error vectors over $\mathbb{F}_{q^m}$ makes the linearization attack possible again. Recall the expression for the ciphertext:

$$\mathbf{r} = \mathbf{m}\mathbf{G} + \sum_{i=1}^{\ell} \alpha_i \boldsymbol{\rho}^{(i)} + \mathbf{e} = (\mathbf{m} + \sum_{i=1}^{\ell} \alpha_i \mathbf{m}'^{(i)})\mathbf{G} + \sum_{i=1}^{\ell} \alpha_i \mathbf{e}'^{(i)} + \mathbf{e}. \quad (22)$$

Let the rank of the error vectors $\mathbf{e}'^{(i)}$'s over $\mathbb{F}_{q^m}$ be denoted by u , i.e., u is the left-hand-side of (21). For an attacker, it is possible to identify u of the $\boldsymbol{\rho}^{(i)}$'s such that their corresponding errors $\mathbf{e}'^{(i)}$ are linearly independent. This can be done, for instance, by finding the $k + u$ linearly independent rows in the matrix

$$\begin{bmatrix} \mathbf{G} \\ \boldsymbol{\rho}^{(1)} \\ \vdots \\ \boldsymbol{\rho}^{(\ell)} \end{bmatrix}. \text{ Without loss of generality, suppose these } u \text{ vectors are } \boldsymbol{\rho}^{(1)}, \dots, \boldsymbol{\rho}^{(u)}.$$

Then, we can rewrite the ciphertext as

$$\mathbf{r} = (\mathbf{m} + \tilde{\mathbf{m}})\mathbf{G} + \sum_{i=1}^u \alpha'_i \boldsymbol{\rho}^{(i)} + \mathbf{e} \quad (23)$$

for unknowns $\alpha'_i \in \mathbb{F}_{q^m}$ and $\tilde{\mathbf{m}} \in \mathbb{F}_{q^m}^k$ is some unknown linear combination of the $\mathbf{m}'^{(i)}$ in the public key. The linearization attack will now succeed with high probability if $n \geq (u + 2)w + k' + u$.⁹ Note this is just (the negation of) condition (12) with ℓ replaced with u . Since u is much smaller, the attack succeeds in recovering $\mathbf{m} + \tilde{\mathbf{m}}$. The vector $\tilde{\mathbf{m}}$ is some linear combination of the vectors $\mathbf{m}'^{(i)}$ and since LIGA uses the shorter message variation, it is possible to unambiguously extract $\mathbf{m}$ from $\mathbf{m} + \tilde{\mathbf{m}}$.

6 Conclusion

To conclude, we will discuss some code families that do not seem to yield a secure instantiation of the framework and share some ideas for future research.

6.1 Insecure Code Classes

Reed-Solomon and Gabidulin codes. As discussed in Section 5.3, both these code classes are insecure in our framework. This is because the linearization attack shows that the corresponding supercode decoding problem is easy when the parameters are chosen such that decryption is possible.

⁹ k' was defined in Section 5.3 and equals $k - \ell$ since LIGA employs the shorter message variant. In fact, in LIGA it is necessary to use this variant since it is not possible to unambiguously extract the α_i 's from $\mathbf{e} + \sum_{i=1}^{\ell} \alpha_i \mathbf{e}'^{(i)}$ during decryption since the $\mathbf{e}'^{(i)}$'s are linearly dependent over $\mathbb{F}_{q^m}$.

BCH and Other Structured Code Classes. Code classes that have a decoding procedure that involves solving a linear system of equations involving the syndrome coefficients (obtained from a “key equation”) are likely to be insecure. This is because the framework can equivalently be formulated in a *parity-check version* (see Appendix C) where the public key and ciphertext are given by

$$\begin{aligned}\boldsymbol{\rho} &= \mathbf{e}'\mathbf{H}^\top, \\ \mathbf{y} &= \mathbf{e}\mathbf{H}^\top + \alpha\boldsymbol{\rho}.\end{aligned}$$

This implies that we can solve the supercode decoding problem by replacing each syndrome coefficient appearing in the linear system of the decoder with a term of the form $y_i - \alpha\rho_i$. Thus, we have only introduced one extra unknown α , and solving the decoding problem will be easy as long as we have enough equations. The latter is almost certainly true since the parameters need to be chosen such that decoding w' erasures and w errors simultaneously is possible, which implies that w cannot be its maximum value. Thus, many structured code classes, including BCH codes, which admit such a decoding procedure are insecure choices in the framework.

Low-Rank Parity Check (LRPC) Codes. LRPC codes [8] are another class of rank metric codes that have an efficient decoder [1] and, in fact, also an error-erasure decoder [10]. The concept of erasures used here corresponds to knowing the column support. We considered these codes in our framework, but the scheme turned out to be insecure. In further detail, we set up the public key and the ciphertext in the parity-check formulation (Appendix C) as

$$\begin{aligned}\boldsymbol{\rho} &= \mathbf{e}'\mathbf{H}^\top, \\ \mathbf{y} &= \mathbf{e}\mathbf{H}^\top + \alpha\boldsymbol{\rho},\end{aligned}$$

where α is drawn from $\mathbb{F}_q^*$ instead of $\mathbb{F}_{q^m}^*$ to preserve the column support of $\mathbf{e}'$ and thus allow Alice to error-erasure decode (using the basic decoding algorithm described in, for example, [1]). Such a scheme can then be attacked as follows: compute the subspace $\mathcal{J} = \langle \mathbf{y} \rangle_{\mathbb{F}_q} + \langle \boldsymbol{\rho} \rangle_{\mathbb{F}_q}$ and the subspace $\mathcal{X} = \cap_i (f_i^{-1} \mathcal{J})$, where $(f_1, \dots, f_d)$ is a basis of the d -dimensional $\mathbb{F}_q$ -space spanned by the entries of the low-rank parity check matrix $\mathbf{H}$ and $a\mathcal{J}$ denotes the subspace $\{ax | x \in \mathcal{J}\}$, where $a \in \mathbb{F}_{q^m}$. Experimentally, it turns out that $\mathcal{X}$ equals the space $\langle \mathbf{e} \rangle_{\mathbb{F}_q}$ with high probability. It is then straightforward to obtain the vector $\mathbf{e}$ by solving a linear system of equations.

6.2 Outlook

If a secure choice of code is possible then, given the above results, we think it is likely that the code family will not have much algebraic structure and would be more combinatorial in nature (LRPC codes notwithstanding). Of course, the code family needs to also have an error-erasure decoder, but for many code families, error-erasure decoding is possible as long as error decoding is possible.

Given the successful attacks outlined above, it is also plausible that no secure code class exists. Nonetheless, our framework is one way to realize an alternative PKE scheme where the secret information to decrypt is a property of the error rather than the code. The idea is natural, and even if not secure in its current form, investigating variations of the framework may lead to secure realizations. For instance, the current method of randomizing the public key by multiplying it with a non-zero field element may be overly simplistic and modifications in this direction are worth exploring.

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A Hard Problem of the Interleaved Variant

The below statement applies to the Hamming metric but the rank-metric version can be easily obtained by replacing the condition $|\cup_{i=1}^{\ell} \text{supp}(\mathbf{e}'^{(i)})| = w'$ with $|\text{supp}(\mathbf{e}'^{(1)}) + \dots + \text{supp}(\mathbf{e}'^{(\ell)})| = w'$ and with the understanding that $\text{supp}(\cdot)$ now denotes the row support.

Problem 2 (Supercode Decoding with Interleaving $(\mathbf{G}, w, w', \ell)$). Draw uniformly at random $\mathbf{c}'^{(1)}, \dots, \mathbf{c}'^{(\ell)} \in \langle \mathbf{G} \rangle$ and $\mathbf{e}'^{(1)}, \dots, \mathbf{e}'^{(\ell)} \in \mathbb{F}_{q^m}^n$ such that

$$|\cup_{i=1}^{\ell} \text{supp}(\mathbf{e}'^{(i)})| = w'.$$

Set $\boldsymbol{\rho}^{(i)} = \mathbf{c}'^{(i)} + \mathbf{e}'^{(i)}$ for $i = 1, \dots, \ell$. Draw $\mathbf{c}^\dagger \in \left\langle \begin{bmatrix} \mathbf{G} \\ \boldsymbol{\rho}^{(1)} \\ \vdots \\ \boldsymbol{\rho}^{(\ell)} \end{bmatrix} \right\rangle \setminus \langle \mathbf{G} \rangle$ and $\mathbf{e} \in \mathcal{S}_w$

uniformly at random. Set $\mathbf{r} = \mathbf{c}^\dagger + \mathbf{e}$. Given only $\mathbf{G}, \boldsymbol{\rho}^{(1)}, \dots, \boldsymbol{\rho}^{(\ell)}$ and $\mathbf{r}$, find $\mathbf{e}$.

B Incompatible Inequalities

Let us recall the three required inequalities on the parameters, which are given by

$$2w + w' \leq n - k, \tag{11}$$

$$n < (\ell + 2)w + k' + \ell, \tag{12}$$

$$w' > \frac{\ell}{\ell + 1}(n - k). \tag{13}$$

The first two inequalities are equivalent, respectively, to

$$\begin{aligned} w &\leq \frac{1}{2}(n - k - w'), \\ w &> \frac{n - \ell - k'}{\ell + 2}. \end{aligned}$$

Together these imply

$$\frac{n - \ell - k'}{\ell + 2} < \frac{1}{2}(n - k - w').$$

and rearranging shows this is equivalent to

$$w' < (n - k) \left(\frac{\ell}{\ell + 2} \right) + 2 \left(\frac{\ell + k' - k}{\ell + 2} \right).$$

The first term on the right-hand side is less than the right hand-side of (13) and the second term is at most 2. Thus, for any parameter set that guarantees meaningful security, this condition contradicts (13).

C Parity-Check Version of Framework

As in the McEliece PKE, there is an equivalent “Niederreiter formulation” of the system which uses the parity-check matrix $\mathbf{H}$ of the public code instead (see Algorithms 7 and 8 below).

Algorithm 7: Keygen (Parity-Check Version)

Input: —
Output: public key $\rho \in \mathbb{F}_{q^m}^{n-k}$,
 secret key $\kappa \in \mathbb{F}_{q^m}^n$
 1 $e' \xleftarrow{\$} \mathbb{F}_{q^m}^n$ s.t. $\text{wt}(e') = w'$
 2 $\rho \leftarrow e' \mathbf{H}^T$
 3 $\kappa \leftarrow e'$
 4 **return** (ρ, κ)

Algorithm 8: Encryption (Parity-Check Version)

Input: message $e \in \mathbb{F}_{q^m}^n$ of
 weight w , public key
 $\rho \in \mathbb{F}_{q^m}^{n-k}$
Output: ciphertext $y \in \mathbb{F}_{q^m}^{n-k}$
 1 $\alpha \xleftarrow{\$} \mathbb{F}_{q^m}^*$
 2 $y \leftarrow e \mathbf{H}^T + \alpha \rho$
 3 **return** y

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